Algebra 1 Name____________________
Unit 1 Date_____________________
Lesson 5 Homework Period____________________

1. In radical form, the expression $x^{\frac{1}{2}}$ is equivalent to ________.

2. In exponential form, the expression $\sqrt[3]{x}$ is equivalent to ________.

For questions 3-5, rewrite each exponential expression as a radical expression. Then, evaluate the expression.

3. $(-125)^{\frac{1}{3}} =$

4. $(1024)^{\frac{1}{5}} =$

5. $49^{\frac{1}{2}} =$

6. Simplify the expression using the laws of exponents and then find the value.

$36^{-\frac{5}{4}} \cdot 36^{\frac{7}{4}} =$

The expression $\frac{81^{\frac{10}{12}}}{81^{\frac{7}{12}}}$ can be rewritten using the law of exponents.

7. Which law of exponents can be used to rewrite the expression?

8. Rewrite the expression as an equivalent expression.

9. Write simplified expression in radical form.

10. Evaluate the radical expression.